Clinical Pilot Proposal

AI-Driven ICU Sepsis Early-Warning Workstation

Collaborative, Privacy-Preserving Clinical Decision Support (FPDAF)

A Pitch for Hospital Partnership, Data Collaboration, and Pilot ICU Deployment

Presented by: **FPDAF Clinical AI Research Team**

Clinical Profile of ICU Sepsis

What Sepsis is

Sepsis is a life-threatening medical emergency caused by the body's extreme, systemic immune response to an infection, which ends up damaging its own tissues, blood vessels, and vital organs.

Why it is Dangerous

If not diagnosed immediately, it rapidly progresses to Septic Shock, leading to multi-organ failure, a sudden drop in blood pressure, and high mortality rates.

The Clinical Time-Window Challenge

Sepsis is notoriously difficult to detect early because its initial symptoms resemble general infections. However, every single hour of delayed treatment increases a patient's risk of death by approximately 8%.

Role of Our Project

By continuously analyzing ICU vital signals, our FPDAF early-warning system acts as an early-warning radar system, alerting doctors hours before a patient clinically deteriorates, giving clinicians a vital head-start to administer antibiotics and fluids.

The Sepsis Challenge in ICU Care

Clinical Bottleneck:

- **High Sepsis Mortality:** Sepsis is a primary cause of ICU mortality. Every hour of delayed diagnosis increases mortality risk by **8%**.
- **Continuous Vitals Overload:** ICU staff are bombarded with raw monitor data, causing alarm fatigue and delayed clinical intervention.
- **Data Silos:** Existing AI tools require centralizing patient records, which is blocked by security policies.

The Compliance & Trust Gap

- **Regulatory Firewalls:** HIPAA and DPDPA (Section 8) regulations strictly prohibit transferring raw patient health data outside hospital firewalls.
- **Black-Box AI Models:** Traditional AI models do not explain *why* an alert is triggered, leading to low adoption by physicians.

The Proposed Solution: FPDAF

What is FPDAF?

The **Federated Personalized Drift-Aware Attention System** (FPDAF) is a decentralized, secure clinical early-warning platform. It trains sequential models collaboratively across multiple clinical centers without ever exchanging raw patient records.

1. Local Security (Zero Data Transfers)

All clinical tensors stay 100% behind your local firewall. Only model weights are aggregated (fully HIPAA & DPDPA compliant).

2. Adaptive Personalization (Ditto Optimization)

Unlike rigid central models, FPDAF uses Ditto optimization to adapt local classifier heads to fit your specific patient demographics.

3. Explainable AI (Bedside Interpretability)

Multi-head temporal self-attentions map risk alerts back to specific patient vitals and ICU stay hours, explaining predictions to physicians.

Clinical Model: Inputs & Outputs

[Input] 1. Our Input (24h Sliding Window)

The input is a rolling 24-hour sequence window of multivariate physiological data for an ICU patient, consisting of:

- **Continuous Vital Signs:** Heart Rate (HR), Systolic Blood Pressure (SBP), Body Temperature, Oxygen Saturation (SpO_2), and Respiration Rate.
- **Sparse Laboratory Measurements:** Hourly blood work values (like WBC, Platelets, Creatinine, and Lactate levels).
- **Static Demographics:** General patient parameters (Age, Gender, and Ward location).

[Output] 2. Our Output / Prediction

The model processes this 24-hour sequence and outputs three specific results:

- **Sepsis Risk Probability Score:** A real-time probability percentage (from 0% to 100%) indicating the sepsis likelihood.
- **Bedside Alarm Alert (Stable vs. Critical):** A binary clinical notification flag triggered if the risk probability score crosses the safety threshold.
- **Temporal Attention Heatmap (XAI):** A timeline overlay showing the exact attention weights computed for each hour, telling the clinician exactly which specific hours/trends (e.g., drop in SpO_2 at hour 15, spike in HR at hour 19) caused the high risk.

How FPDAF Helps Clinicians (Doctor Portal)

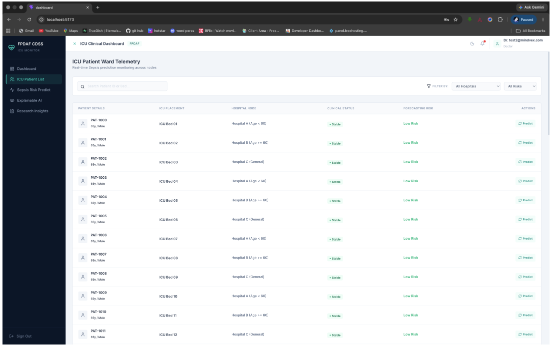

Figure 1: Doctor's Bedside Diagnostic View.

Key Clinical Support Workflows

- **Real-Time Warning Dials:** Visualizes sepsis risk indices hours before clinical shock or physical collapse.
- **Explainable Attributions:** Displays exactly which hours in the last 24h (e.g. heart rate spike at hour 14) influenced the alert.
- **Hourly Vitals Overlay:** Integrates HR, SBP, Temp, Resp, and SpO_2 timelines into a unified interactive monitor.
- **Reduced Alarm Fatigue:** Calibrates alerts locally to prevent false positive notifications.

How FPDAF Helps Management (Admin & Security)

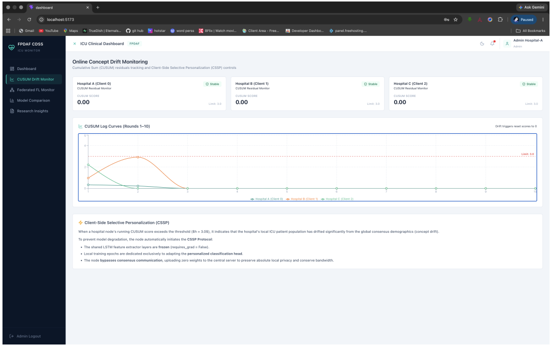

Figure 2: Admin's ML & Drift Telemetry View.

Institutional & Compliance Benefits

- **100% HIPAA & DPDPA Compliant:** Zero raw patient data leaves your data center.
- **Statistical Drift Audit (CUSUM):** Monitors local validation residuals to alert IT staff of demographic shifts in real-time.
- **Bandwidth Savings (CSSP):** Freezes deep backbones during stable phases, cutting network overhead by **38%** vs. standard personalization.
- **Low Compute Burden:** Local model head fine-tuning completes in **6 minutes** on standard hospital hardware.

What We Need: Collaborative Partnership Proposal

We are seeking progressive healthcare institutions to partner in a pilot study:

We do not require budget allocations. We are proposing a technical collaboration consisting of three items:

1. Clinical Guidance (Physician Feedback)

Access to ICU physicians to review the bedside UI, validate alert thresholds, and provide clinical input on explainability mappings.

2. Local Docker Client (IT Infrastructure)

Permission to host a secure, dockerized local model client inside your data center. The server aggregates only weight tensors (low compute).

What We Need: Required De-Identified Data

3. Retrospective Validation (Required ICU Telemetry & Clinical Records)

To tune the local personalized classifier head to your patient demographics and run retrospective validation trials, we require access to historical, de-identified ICU patient records containing:

- **Hourly Vital Signs:** Heart Rate (HR), Systolic/Diastolic/Mean Blood Pressure, Respiration Rate, Body Temperature, and Oxygen Saturation (SpO_2).
- **Laboratory Measurements:** Routine blood panels (White Blood Cell count, Platelets, Creatinine, Bilirubin, Blood Urea Nitrogen, Glucose, and Lactate levels).
- **Clinical Outcome Labels:** Time of Sepsis onset (defined by Sepsis-3 clinical criteria) or ICD-10 diagnostic discharge codes.
- **De-Identification Standards (Strict Zero PII):** All records must be pre-stripped of patient names, medical record numbers, phone numbers, exact zip codes, and specific timestamps to maintain 100% HIPAA and DPDPA compliance.

Roadmap to Clinical Pilot Deployment

Phase 1: Validation
Clinical interface review & data mapping.
(1-2 Weeks)

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Phase 2: Offline Tuning
Host local Docker client & evaluate historical data.
(3-4 Weeks)

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Phase 3: Silent Run
Run model in parallel with clinical workflow.
(4 Weeks)

Goal of the Pilot Program

Evaluate the early-warning accuracy of FPDAF in parallel with daily clinical operations, with zero risk to patients, in order to prove diagnostic utility and safety.

Thank You

Questions & Discussion

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